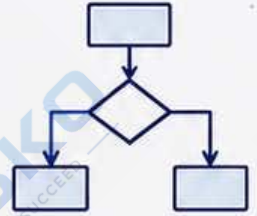


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Question Bank



Data Science

Unit I - Introduction to Data Science

- **Answers in one or two lines (1 mark questions)**

1. Define Data Science.
2. What is the importance of Data Science?
3. What is the Data Science Lifecycle?
4. Name the major stages of the Data Science Lifecycle.
5. What is data?
6. What is structured data?
7. What is unstructured data?
8. What is semi-structured data?
9. Name any two types of data.
10. What is Big Data?
11. What is Artificial Intelligence?
12. Expand AI.
13. Name any two applications of Data Science.
14. Name any two Data Science tools.
15. Name any two Data Science platforms.
16. What is data analysis?
17. What is data visualization?
18. What is predictive analysis?
19. What is the role of Python in Data Science?
20. What is the difference between Data Science and Artificial Intelligence?

● **Short Answer Questions (3 marks questions)**

1. Define Data Science and explain its importance.
2. Explain the Data Science Lifecycle.
3. Explain the major stages of the Data Science Lifecycle.
4. Explain the applications of Data Science.
5. Explain different types of data.
6. Differentiate between structured, semi-structured, and unstructured data.
7. Explain the importance of data in Data Science.
8. Explain any three Data Science tools.
9. Explain any three Data Science platforms.
10. Explain the role of Python in Data Science.
11. Explain the concept of Big Data.
12. Explain the main characteristics of Big Data.
13. Explain Artificial Intelligence and its relationship with Data Science.
14. Differentiate between Data Science and AI.
15. Explain how Data Science is used in real-life applications.

● **Long Answer Questions (6 marks questions)**

1. Define Data Science and explain its importance, scope, and applications in detail.
2. Explain the Data Science Lifecycle with all its stages and a suitable diagram.
3. Explain different types of data with suitable examples.
4. Differentiate between structured, semi-structured, and unstructured data with examples.
5. Explain various applications of Data Science in real-life fields.
6. Explain important Data Science tools and platforms with their uses.
7. Explain Big Data, its characteristics, and its relationship with Data Science.
8. Explain Artificial Intelligence and its relationship with Data Science.
9. Explain the role of Python and other tools in the Data Science process.
10. Explain the complete Data Science Lifecycle with suitable examples from a real-world problem.

Unit II - Data Collection and Preprocessing

- **Answers in one or two lines (1 mark questions)**

1. What is Data Collection?
2. Define Data Preprocessing.
3. What is structured data?
4. What is unstructured data?
5. Name any two methods of data collection.
6. What is data cleaning?
7. What is a missing value?
8. What is data transformation?
9. Define data normalization.
10. What is data integration?
11. What is data reduction?
12. Why is data preprocessing required?
13. What is duplicate data?
14. What is noisy data?
15. What is an outlier?
16. Name any two techniques for handling missing values.
17. What is the purpose of normalization?
18. What is the role of Python in Data Science?
19. Name any two Python libraries used in Data Science.
20. What is NumPy?

● **Short Answer Questions (3 marks questions)**

1. Explain different methods of data collection.
2. Differentiate between structured and unstructured data.
3. Explain the importance of data preprocessing.
4. Explain data cleaning with suitable examples.
5. Explain different methods of handling missing values.
6. Explain duplicate and noisy data.
7. Explain outliers in a dataset.
8. Explain data transformation.
9. Explain data normalization.
10. Explain the need for data integration.
11. Explain data reduction techniques.
12. Explain the steps involved in data preprocessing.
13. Explain the role of Python in Data Science.
14. Explain NumPy and its applications in Data Science.
15. Explain any three Python libraries commonly used in Data Science.

● **Long Answer Questions (6 marks questions)**

1. Explain Data Preprocessing and its major steps in detail.
2. Explain different data collection methods with suitable examples.
3. Explain structured and unstructured data with suitable examples.
4. Explain data cleaning and different techniques for handling missing values.
5. Explain data transformation and normalization with suitable examples.
6. Explain data integration and data reduction in detail.
7. Explain different problems in raw data such as missing values, duplicates, noise, and outliers, and describe methods to handle them.
8. Explain the complete data preprocessing process with a suitable example.
9. Explain the role of Python in Data Science and discuss important Python libraries used for data analysis.
10. Explain data cleaning, transformation, normalization, integration, and reduction as preprocessing techniques.

Unit III - Risk Management and Design Engineering

- **Answers in one or two lines (1 mark questions)**

1. Define Data Visualization.
2. What is the importance of Data Visualization?
3. What is a chart?
4. What is a graph?
5. Define Bar Chart.
6. What is a Pie Chart?
7. Define Histogram.
8. What is a Scatter Plot?
9. What is Descriptive Statistics?
10. Define Mean.
11. Define Median.
12. Define Mode.
13. What is Correlation?
14. What is Regression?
15. What is Probability?
16. What is a positive correlation?
17. What is a negative correlation?
18. What is zero correlation?
19. What is the range of probability?
20. Name any two measures of central tendency.

- **Short Answer Questions (3 marks questions)**

1. Explain the importance of Data Visualization in Data Science.
2. Explain Charts and Graphs.
3. Explain Bar Chart with a suitable example.
4. Explain Pie Chart with a suitable example.
5. Explain Histogram with a suitable example.
6. Explain Scatter Plot with a suitable example.
7. Differentiate between Bar Chart and Histogram.
8. Differentiate between Histogram and Scatter Plot.

9. Explain Descriptive Statistics.
10. Explain Mean, Median, and Mode.
11. Differentiate between Mean, Median, and Mode.
12. Explain the concept of Correlation.
13. Explain positive, negative, and zero correlation.
14. Explain the basics of Regression.
15. Explain Probability and its importance in Data Science.

● **Long Answer Questions (6 marks questions)**

1. Explain Data Visualization and its importance in Data Science with suitable examples.
2. Explain Bar Chart, Pie Chart, Histogram, and Scatter Plot with suitable diagrams.
3. Differentiate between Bar Chart, Pie Chart, Histogram, and Scatter Plot.
4. Explain Descriptive Statistics and its important measures with suitable examples.
5. Explain Mean, Median, and Mode with formulas and suitable numerical examples.
6. Explain Correlation and its different types with suitable examples and diagrams.
7. Explain the basics of Regression and its importance in Data Science.
8. Explain Probability concepts used in Data Science with suitable examples.
9. Explain the relationship between Correlation and Regression.
10. Explain how Data Visualization and Statistics are used together in Data Science with a suitable example.

Unit IV - Introduction to Machine Learning

- **Answers in one or two lines (1 mark questions)**

1. Define Machine Learning.
2. What is Machine Learning?
3. What is the basic idea behind Machine Learning?
4. Define Supervised Learning.
5. Define Unsupervised Learning.
6. What is Classification?
7. What is Regression in Machine Learning?
8. What is Clustering?
9. What is a training dataset?
10. What is a testing dataset?
11. What is a feature in Machine Learning?
12. What is a label?
13. Give one example of Supervised Learning.
14. Give one example of Unsupervised Learning.
15. Name any two applications of Machine Learning.
16. What is a classification problem?
17. What is a regression problem?
18. What is meant by ethical issues in Data Science?
19. What is data privacy?
20. What is bias in Machine Learning?

● **Short Answer Questions (3 marks questions)**

1. Explain Machine Learning and its importance.
2. Explain the basic Machine Learning process.
3. Explain Supervised Learning with examples.
4. Explain Unsupervised Learning with examples.
5. Differentiate between Supervised and Unsupervised Learning.
6. Explain Classification with a suitable example.
7. Explain Regression with a suitable example.
8. Differentiate between Classification and Regression.
9. Explain the basic concept of Clustering.
10. Explain the difference between Classification and Clustering.
11. Explain the role of training and testing data in Machine Learning.
12. Explain the applications of Machine Learning in real life.
13. Explain Machine Learning applications in healthcare and finance.
14. Explain ethical issues in Data Science.
15. Explain data privacy, security, and bias in Machine Learning.

● **Long Answer Questions (6 marks questions)**

1. Explain Machine Learning, its basic concepts, and its importance in Data Science.
2. Explain Supervised and Unsupervised Learning with suitable examples.
3. Differentiate between Supervised and Unsupervised Learning in detail.
4. Explain Classification and Regression with suitable examples.
5. Differentiate between Classification and Regression with suitable examples.
6. Explain Clustering and its basic concept with suitable examples.
7. Explain the major applications of Machine Learning in real-life situations.
8. Explain ethical issues in Data Science and Machine Learning in detail.
9. Explain data privacy, security, bias, fairness, and transparency as ethical issues in Data Science.
10. Explain the different types of Machine Learning and their real-life applications.

